

Exercise 6: Implementing the Proxy Pattern

CODE:

```
// Subject Interface

interface Image {

    void display();

}

// Real Subject

class ReallImage implements Image {

    private String fileName;

    ReallImage(String fileName) {

        this.fileName = fileName;

        loadFromServer();

    }

    private void loadFromServer() {

        System.out.println("Loading " + fileName + " from remote server...");

    }

    public void display() {

        System.out.println("Displaying " + fileName);

    }

}

// Proxy Class

class ProxyImage implements Image {

    private String fileName;

    private ReallImage reallImage;

    ProxyImage(String fileName) {

        this.fileName = fileName;

    }

    public void display() {

        if (reallImage == null) {
```

```
        reallImage = new RealImage(fileName);
    }
    reallImage.display();
}
}
// Main Class
public class Main {
    public static void main(String[] args) {
        Image image = new ProxyImage("Library.jpg");
        System.out.println("First Display");
        image.display();
        System.out.println();
        System.out.println("Second Display");
        image.display();
    }
}
```

OUTPUT:

```
First Display
Loading Library.jpg from remote server...
Displaying Library.jpg

Second Display
Displaying Library.jpg
```